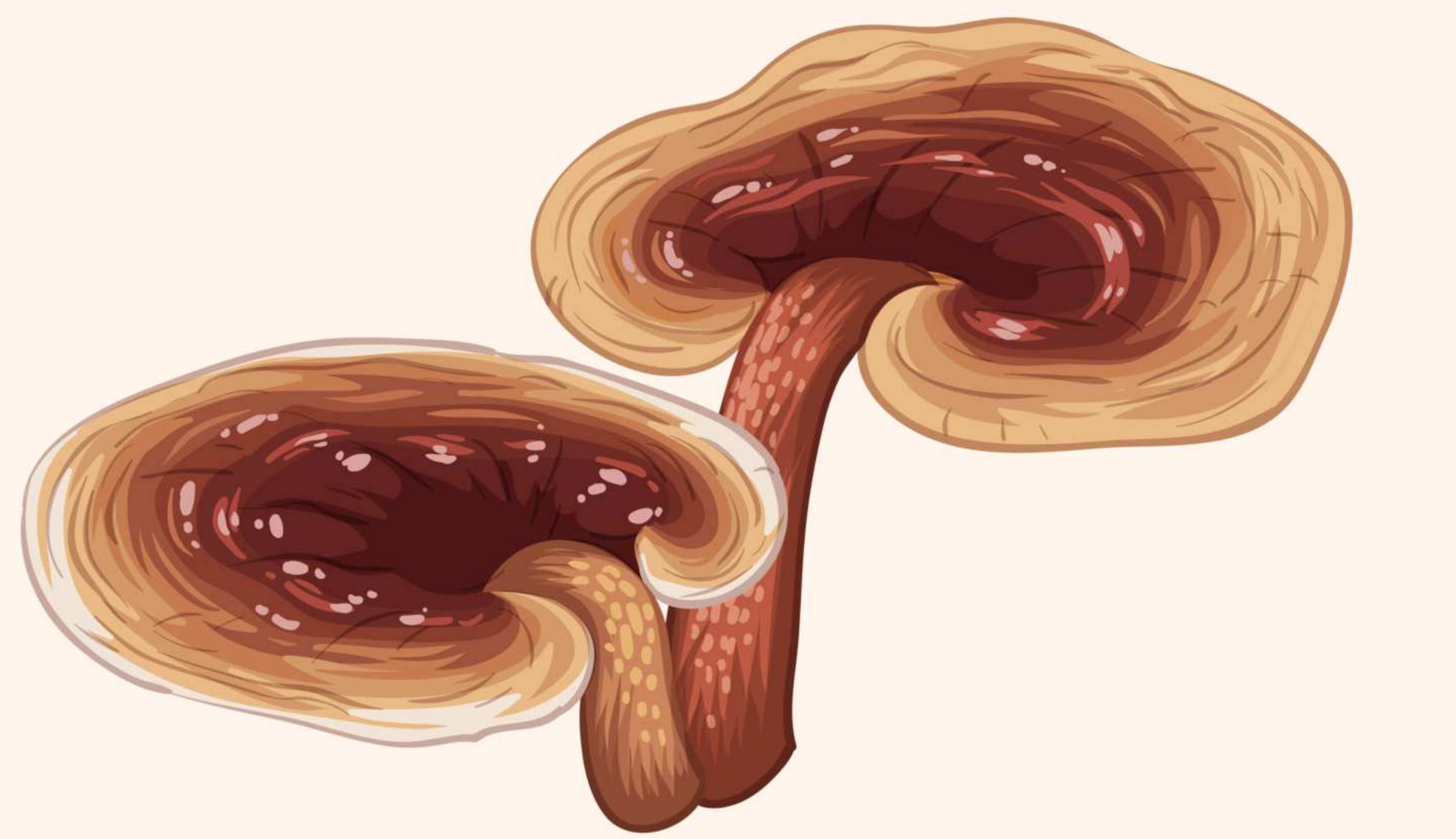




AZ Elements. (n.d.). Shiny Reishi Mushroom Still Life [Graphic]. In *Canva Pro*. www.canva.com

Background

With rapidly growing populations and pollution, sub-Saharan African countries are in need of **cheap and environmentally-friendly alternatives to traditional construction materials**. Mycelium bricks are an example of these materials and have potential in temporary **construction applications**; they are made of **low-cost agricultural waste substrates inoculated by fungi** (Camilleri et. al, 2025). Common agriculture waste products in sub-Saharan Africa are **sugarcane bagasse (SB)**, **coconut coir (CC)**, and **rice husks (RH)** with biopolymers for energy (Zoghلامي & Paes, 2026).



Figure 1. Image of baked mycelium brick

<https://happho.com/an-emerging-sustainable-construction-material-mycelium-bricks/>

Purpose

- Mycelium bricks are **weaker than their traditional counterparts**, potentially making them inviable in real life (Camilleri et. al, 2025). By testing varying **weight concentrations of SB, CC, and RH in substrate mixes**, the goal was to increase their compressive strength. These materials have not been tested together as substrates for mycelium bricks before. **It was hypothesized that substrates of a higher weight concentration of RH would have the highest compressive strength** since RH are renowned for its high porosity, surface area, and aeration, all vital for fungi (Poffa Costa et. al, 2023).
- Ganoderma lucidum* was the fungi of choice since it is renowned for its **fast growth and structural superiority** (Alemu et al., 2022). The brick-making process was designed to be as simple as possible for **rural regions** that lack specialized resources.

Substrate Preparation

Inoculation and Molding

Strength Testing

Determining the Effects of Various Substrate Mixes of Sugarcane Bagasse, Coconut Coir, and Rice Husks on the Compressive Strength of *Ganoderma lucidum* Mycelium Bricks

Materials Science and Design

Methods

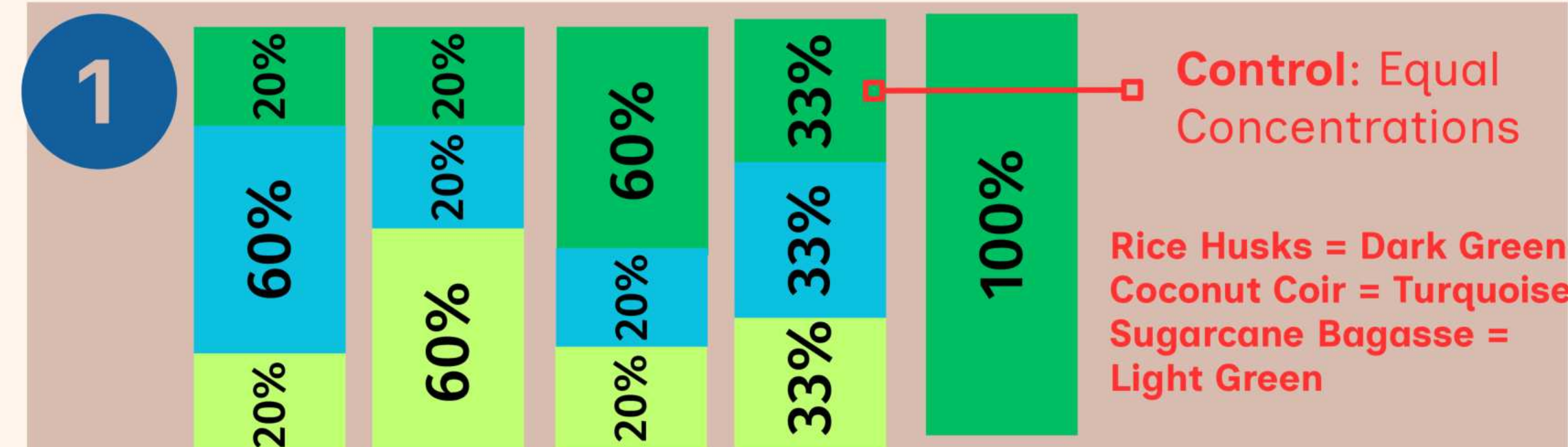


Figure 2. Bagasse, rice husks, and coconut coir concentration groups

- Sugarcane bagasse plates cut into ~1 in pieces and coconut coir bricks broken apart into fibers.
- 300g of each substrate concentration prepared in bins and moisturized with 450 - 600 mL of water



Mycelial growth on surface

Figure 5. Tray of growing mycelium brick replicates as of 5/15/2026

Photos taken by student researcher



Figure 6. 60% SB brick of complete inoculation after 50 days

- Inoculated substrate mix broken up and added to 6 brick replicates (8.00 x 5.46 x 2.03 cm) and placed into incubator for final colonization (still ongoing)
- Remaining 60% SB substrate mix continued to inoculate; others stagnated with no cohesive structure

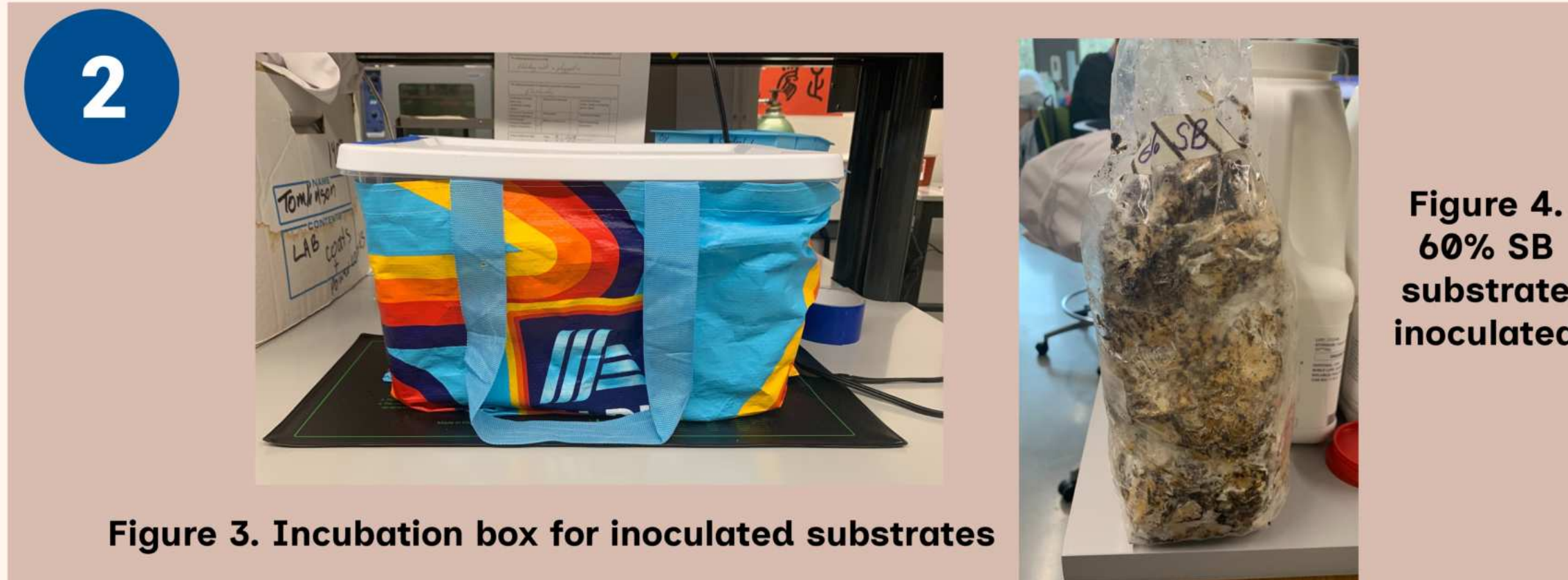


Figure 3. Incubation box for inoculated substrates

Photos taken by student researcher

- Substrates autoclaved and inoculated with 3 mL of culture in bags
- Bags transferred to incubation setup over 31 days

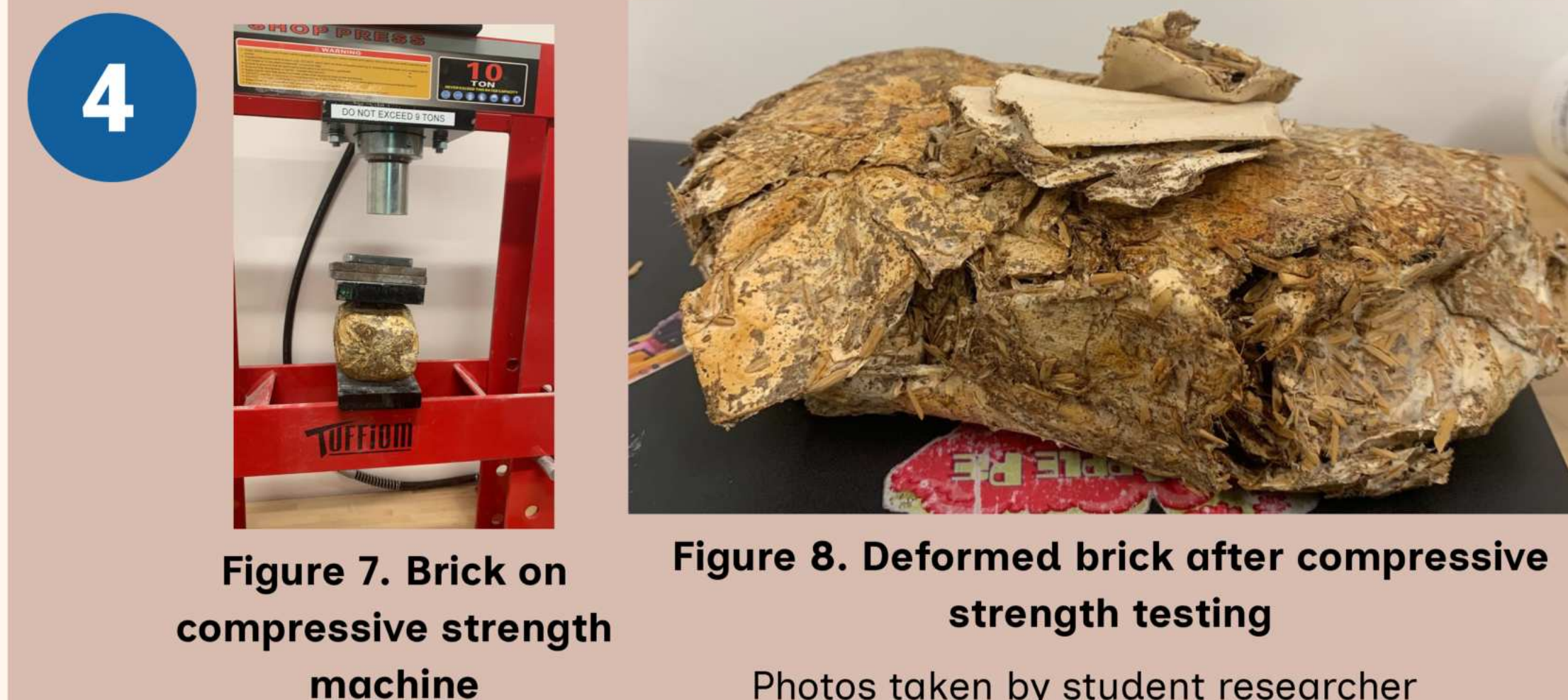


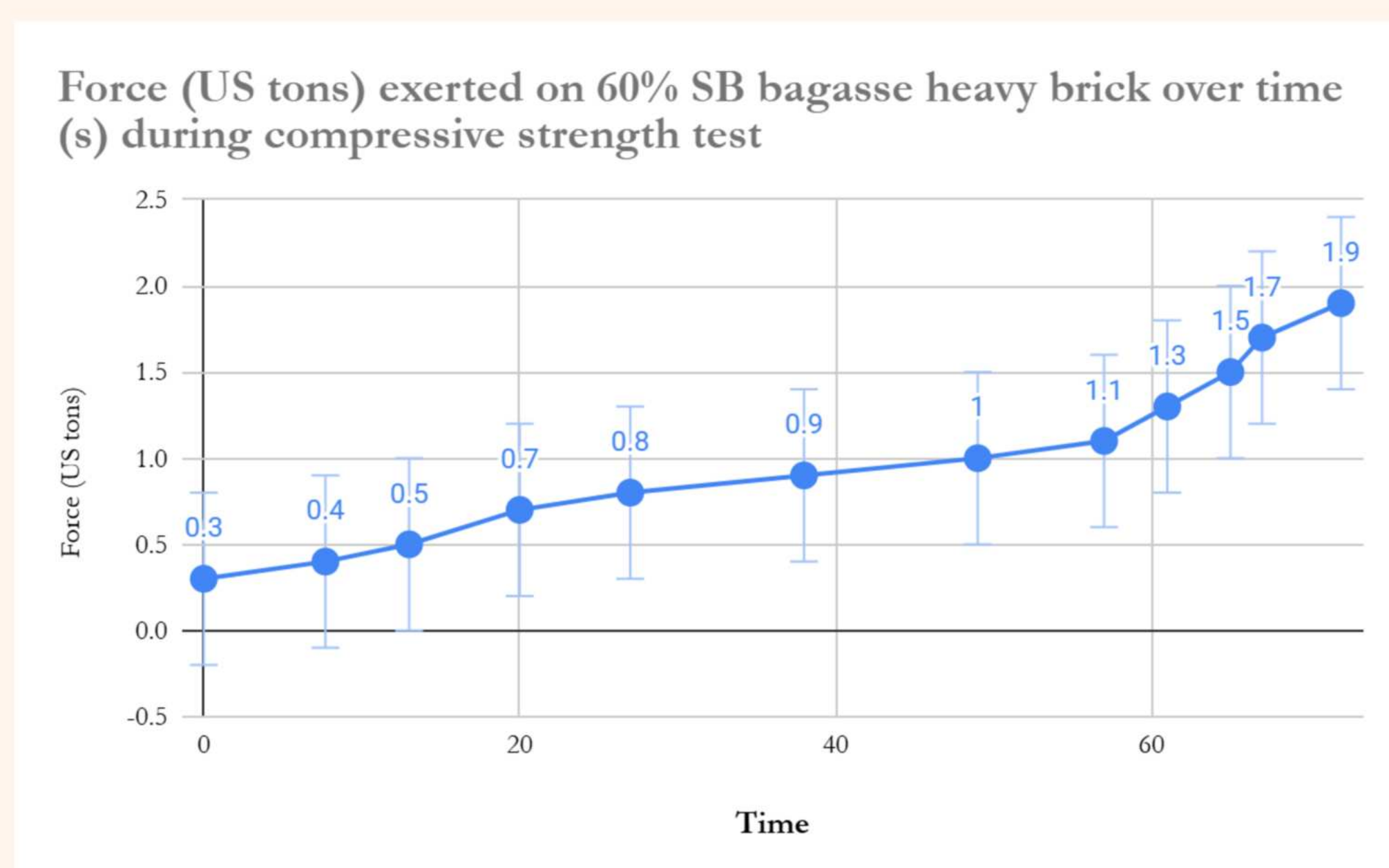
Figure 7. Brick on compressive strength machine

Figure 8. Deformed brick after compressive strength testing

Photos taken by student researcher

- 60% SB brick dried in oven
- Density and moisture content recorded
- Brick tested against increasing load with compressive strength test until significant deformation was reached (to be done with smaller replicates)

Results



Graph 1. Force over time graph from 60% SB brick compressive strength test

Graphs created by student researcher

Force (US tons) exerted on bagasse heavy block over time (s) during compressive strength test	
Time (s) +/- 0.05	Force (US tons) +/- 0.5
0	0.3
7.7	0.4
13	0.5
20	0.7
27	0.8
38	0.9
49	1
57	1.1
61	1.3
65	1.5
67	1.7
72	1.9

Table 1. Force over time table from 60% SB strength test

Table 1. Data table of estimated physical properties of 60% SB brick

Physical Property	60% SB Brick
Density (kg/m ³)	350
Compressive strength (MPa)	1.8
Amount of deformation from testing	40.00%

Compressive strength is the maximum force load exerted on the brick divided by the cross section area.



Husen, Multasam. (n.d.). Two halves of a coconut [Graphic]. In *Canva Pro*. www.canva.com

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Discussion + Future Research

Table 2. Data table of typical mycelium brick properties from Alaneme et. al (2023)

Density (kg/m ³)	110 - 330
Compressive Strength (MPa)	0.1 - 1.0

- The SB dried brick was of estimated mechanical parameters that exceeded **typical density and compressive strength values for mycelium bricks**. The larger surface area of individual SB pieces may have helped with mycelial network formation.
- Contamination and stalled growth**, making the 100% RH and equal concentration substrates unusable for any replicates. Significant **mycelial growth has been observed on smaller replicates but all mycelial networks are too weak for compressive strength testing** which suggests the substrates' properties are not ideal for future mycelium bricks applications.
- There were a small amount of replicates due to limited space and time but future research with **larger replicates of greater quantities** could give more conclusive results.
- Improved areas of incubation with **standard temp. and humidity** rather than fluctuating ranges may allow for improved inoculation.

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